W. D. GANN ASTRONOMICAL HARMONICS & INTRADAY MIDPOINT MANUAL

A Master Reference Guide for Quantitative Trading in Indian Equities, Options & Commodities (NSE, BSE, MCX)

Target Audience: Quantitative Analysts & Professional Traders	Primary Engine: NASA JPL Horizons + Python
Market Coverage: NSE Nifty 50, Bank Nifty, BSE Sensex, MCX	Timezone Standard: Indian Standard Time (IST, UTC+05:30)
App 1 (Astro Ephemeris): gann_timer/ui.py	App 2 (Midpoint Calculator): gann_midpoint_app.py

1. Foundational Concept: Why Astro-Harmonics & Midpoints Work

Legendary market master W. D. Gann stated that market oscillations are governed by the Law of Vibration. Price and time are not independent entities—they are two interchangeable manifestations of circular cycles. Market pivot points occur when circular time geometric divisions align with natural celestial periodicities.

In modern quantitative terms, these tools operate as phase-synchronized volatility oscillators. Standard clock time is human-constructed and ignores Earth's axial progression, whereas institutional order flows, algorithmic volume spikes, and seasonal capital allocation consistently align with astronomical milestones (e.g., Solar Equinoxes, lunar cycle culminations, and intraday half-session boundaries).

> *Gann's Master Axiom: 'When time and price square, a change in trend is inevitable. The 50% point or half-way point of any movement, range, or cycle is the strongest resistance and support point.'*

2. Application 1: Gann Celestial Tracker (Ephemeris Engine)

File: d:\ephemeris\main.py (or d:\ephemeris\Launch_GannCelestialTracker.lnk)
Purpose: Tracks planetary orbital mechanics to generate long-term, swing, and intraday market entry windows synchronized with Indian market operational trading sessions.

2.1 Celestial Phenomena & Market Implications

Phenomenon	Symbol & Angle	Market Effect & Strategy
Conjunction	☌ (0°)	Cycle Genesis: Major trend reversal or consolidation breakout. Heavy institutional repositioning.
Opposition	☍ (180°)	Cycle Polarity: Market exhaustion, counter-trend pivots, blow-off tops or panic selling troughs.
Square	□ (90° / 270°)	Disharmony / Clash: Violent intraday volatility, sudden trend shakes, false breakout traps.
Trine	△ (120° / 240°)	Harmonic Flow: Strong trend acceleration, momentum continuations, smooth trending channels.
Octiles (1/8)	∠ (45°) / ⚼ (135°)	Minor Turning Points: Short-term counter-swings, weekly option expiry pin targets.
Planetary Stations	Stat Rx / Stat Dir	Zero-Velocity Inflections: Strongest macro reversals. Outer planets (Jupiter/Saturn) form multi-week swing pivots.
Cardinal Ingress	0° Ar, Cn, Li, Cp	Gann Seasonal Turns: Solstices & Equinoxes dictate seasonal capital rotation across equity sectors.

2.2 Bidirectional Price ↔ Longitude Conversion Mathematics

The application incorporates Gann's authentic mathematical formulas converting planetary ecliptic degrees into exact market price levels:

EQUATION: SQUARE OF NINE FIXED WHEEL RUNGS & QUADRATIC ROOT HARMONICS

$$P(k, \theta) = [\, 2k + 1.0 + (\theta \bmod 360°) / 180° \,]^2, \; k = \lfloor (\sqrt{P_ref} - 1.0 - \theta / 180°) / 2 \rfloor$$

- Odd integer squares anchor the 0° base axis: $\sqrt{P} = 2k + 1$ (1, 9, 25, 49, 81...)
- Fixed Support / Resistance: Rungs bracketing spot price P_ref along planetary ray θ
- Relative Quadratic Projections: $P_\pm\theta = (\sqrt{P_ref} \pm \theta / 180°)^2$ [+90°: +0.50, +120°: +⅔, +180°: +1.00]

Natural 360° Wheel: Price Modulo 360 equates points directly to circular coordinates (e.g., Nifty at 24,350 = 230° Scorpio).

2.3 Strict Indian Market Rules & Exclusion Compliance

• Session Alignment: Celestial events occurring outside regular market hours (e.g., 03:45 IST) are automatically mapped to the Pre-Open (09:00 or 09:15 IST) of the next trading day.
• Nifty 50 Options Expiry: Accurately calibrated to weekly Tuesday expiries (effective SEBI mandate).
• Critical Exclusion Filter: All Gold, Silver, and Bullion contracts are permanently hard-filtered across all data routing and reports.

3. Application 2: Gann Midpoint & 360° Time Harmonics

File: d:\ephemeris\gann_midpoint_app.py
Purpose: Calculates the exact 50% equilibrium center of any trading day, projecting bidirectional harmonic time angles (Left = Morning, Right = Afternoon) and corresponding 50% price support/resistance levels.

3.1 Core Intraday Mathematical Formulations

EQUATION: 360° SESSION TIME HARMONICS & METHOD 6 PRICE OCTAVES

$$T_{Left}(-\theta) = T_{mid} - [\ \theta \times (Duration / 360°)\ ],\quad T_{Right}(+\theta) = T_{mid} + [\ \theta \times (Duration / 360°)\ ]$$

- Session Time Scales: NSE Equity (375m) 1° = 1.0417m (62.5s) | MCX Commodity (870m) 1° = 2.4167m (145s)
- Method 6 Linear Range Octaves: $P_{Range}(\pm\theta) = P_{50} \pm (\theta / 360°) \times Range = P_{50} \pm (\theta / 180°) \times (Range / 2)$
- Method 6 Sq9 Quadratic Roots: $P_{Sq9_Sup}(-\theta) = [\ \max(0, \sqrt{P_{50}} - \theta/180°)\ ]^2$, $P_{Sq9_Res}(+\theta) = [\ \sqrt{P_{50}} + \theta/180°\ ]^2$
- NOAA Spencer True Solar Noon: $EoT(\Gamma) = 229.18 \times [0.000075 + 0.001868 \cos\Gamma - 0.032077 \sin\Gamma - \ldots]$, Noon = 12:00 + 38.7m − EoT

3.2 The 6 Calculation Methods Explained

Method	Midpoint Formula	Degree Scale	Trading Application
Method 1: NSE/BSE Session	09:15 + 187.5m = 12:22:30 IST	1° = 1.0417 min (62.5 sec)	Primary intraday timing tool for Nifty, Bank Nifty, and stock options. Reversals frequently trigger within ±15 min of 12:22 IST.
Method 2: MCX Session	09:00 + 435m = 16:15:00 IST	1° = 2.4167 min (145.0 sec)	For Crude Oil, Natural Gas, and Base Metals. Shifts to 16:27 IST during US Standard Time. Marks European/US overlap shift.
Method 3: Solar Noon	Mumbai Zenith = 12:22 - 12:54 IST	1° = 4.0000 min (240.0 sec)	Astronomical True Solar Noon corrected for NOAA Spencer Equation of Time. Aligns within seconds of NSE Midpoint.
Method 4: Opening Range	09:15 + 30m = 09:45:00 IST	1° = 10.0 sec (0.1667 min)	First 60-minute range equilibrium (60m = 360°). Retests of the 09:45 midpoint determine morning trend continuation.
Method 5: Custom Pivot Time	User Selected (e.g., 10:45 IST)	Selectable (1.04m, 4m, 10s)	Anchors the 0° balance line at the exact minute an extreme intraday High or Low was registered.
Method 6: Price Equilibrium	(High + Low) / 2 (e.g. 24,350 pts)	Dual: Range 8ths + Sq9 √P Frequencies	Dual Price & Time Squaring: Generates exact numerical price support/resistance levels mapped alongside time degrees.

3.3 Harmonic Degrees Timetable Reference (NSE Equity Session)

Degree	Left (Morning)	Right (Afternoon)	Significance & Strategy Execution
0° (MID)	**□ 12:22:30 IST**	**□ 12:22:30 IST**	THE MIDPOINT OF THE DAY: Trend gravity axis. Price above = Long, Price below = Short.
30°	11:51:15 IST	12:53:45 IST	Semi-Sextile: Pre-midpoint compression and 1:00 PM European market opening reaction.
45°	11:35:37 IST	13:09:22 IST	Octile: High-probability swing reversal window. Frequently caps morning rally or slide.
90°	10:48:45 IST	13:56:15 IST	Major Square: 10:48 terminates initial morning expansion; 13:56 launches afternoon directional wave.
120°	10:17:30 IST	14:27:30 IST	Major Trine: Peak morning liquidity; 14:27 initiates final power-hour momentum surge.
144°	09:52:30 IST	14:52:30 IST	Gann Master 144: Opening range completion and pre-closing square-off window.
180°	09:15:00 IST	15:30:00 IST	Opposition / Extremes: Official trading bell boundaries of the 375-minute day.

4. Step-by-Step Practical Trading Workflow

To achieve maximum edge, professional Gann traders execute a structured 4-phase daily workflow:

Phase 1: Pre-Market Astro Preparation (08:30 - 09:00 IST)

1. Open Gann Celestial Tracker (main.py).
2. Select your target instrument (e.g., Index (NSE/BSE) or Nifty 50 Options).
3. Enter today's date and active reference price (yesterday's close, e.g. 24,350).
4. Click 'Generate Gann Timeline'. Review the celestial triggers logged for today.
5. Note any High Strength aspects or ingresses occurring during market hours. Take note of the calculated Sq9 Support and Resistance price levels.

Phase 2: Midpoint Setup & Degree Projection (09:15 - 10:30 IST)

1. Open Gann Midpoint App (gann_midpoint_app.py).
2. For general index trading, choose Method 1 (NSE/BSE Session). The app locks the midpoint at 12:22:30 IST.
3. If an extreme high or low forms in the first 45 minutes, switch to Method 6 (Price Equilibrium) and enter the Day High and Day Low.
4. Click 'Recalculate Price Octaves'. The table now displays the exact price rungs and timing nodes side-by-side.
5. Set alarms at key upcoming angles: 90° (10:48 / 13:56 IST) and 45° (11:35 / 13:09 IST).

Phase 3: The 12:22 IST Midpoint Execution Rules

Between 12:15 PM and 12:30 PM IST, monitor the market closely:
• Bullish Stance: If Nifty is trading above the 50% price equilibrium and prints a higher-low consolidation into 12:22 IST, initiate long positions targeting the 75% resistance quartile and 90° right time node (13:56 IST).
• Bearish Stance: If Nifty is trading below the 50% price equilibrium, initiate short positions targeting the 25% support quartile.
• Reversal Trigger: If price touches an active Square of Nine resistance level at the same time the clock strikes 12:22 IST, a master price-time squaring has occurred. Look for an immediate trend reversal.

Phase 4: Afternoon Momentum Acceleration (13:56 - 15:30 IST)

1. At 13:56 IST (90° Square), expect afternoon directional breakout momentum.
2. At 14:27 IST (120° Trine), trail stop-losses closely as final daily institutional square-off begins.
3. Export reports to CSV using the built-in 'Export CSV' button to maintain an institutional trade audit journal.

Summary Checklist for Active Traders:
□ Check NASA JPL ephemeris for planetary aspects during session hours.
□ Map the 50% price midpoint of the opening range as your bullish/bearish line of control.
□ Watch the 12:22:30 IST session midpoint for trend exhaustion or secondary acceleration.
□ Trade in the direction of the 50% equilibrium toward the 90° (13:56 IST) and 120° (14:27 IST) time targets.
□ Always confirm Gann astro-windows with volume and disciplined stop-loss risk management.